PHYSICS & ELECTRONICS PROJECT REPORT

Physics & Working Principles of the PN Junction Diode

A Comprehensive Study on Semiconductor Fundamentals, Depletion Dynamics, and Rectification Applications

SUBJECT	DOCUMENT TYPE	TOPIC	STATUS
Semiconductor Physics	**Technical Report**	**PN Junction Diode**	**Final Submission**

1. INTRODUCTION

A **PN junction diode** is one of the most fundamental semiconductor devices in modern electronics. It is formed by joining a p-type semiconductor (rich in positive holes) with an n-type semiconductor (rich in negative electrons) within a single continuous crystal structure. The interface between these two regions is called the **PN junction**. The diode allows electrical current to flow easily in one direction (forward bias) while blocking current in the opposite direction (reverse bias), a property known as *rectification*.

2. FORMATION OF THE DEPLETION REGION & POTENTIAL BARRIER

When a p-type and n-type semiconductor are joined together, a sharp gradient in charge carrier concentration exists across the boundary:

- **Diffusion Process:** Majority electrons from the n-side diffuse across the junction into the p-side, where they recombine with holes. Simultaneously, majority holes from the p-side diffuse into the n-side and recombine with electrons.
- **Space Charge Region (Depletion Layer):** As electrons leave the n-region, they leave behind fixed positive donor ions. As holes leave the p-region, they leave behind fixed negative acceptor ions. This leaves a region near the junction free of mobile charge carriers, termed the *depletion region*.
- **Built-in Potential (Barrier Potential V_0):** The fixed ionic charges establish an internal electric field directed from the n-side to the p-side. This field opposes further diffusion and eventually establishes thermal equilibrium. The voltage difference across the depletion layer is the built-in barrier potential ($V_0 \approx 0.7$ V for Silicon, $V_0 \approx 0.3$ V for Germanium at 300 K).

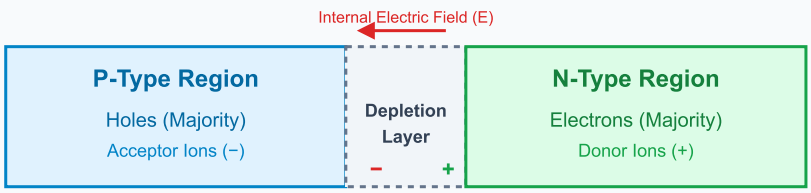


Figure 1: Schematic representation of an unbiased PN junction showing the depletion layer and built-in electric field.

3. BIASING CONDITIONS OF THE DIODE

Biasing refers to applying an external voltage across the terminals of the diode to alter its internal potential barrier.

3.1 Forward Bias Mode

When the positive terminal of an external DC source is connected to the p-type region and the negative terminal to the n-type region:

- The external electric field opposes the internal built-in field.
- The potential barrier height is reduced to $(V_o - V_f)$, and the width of the depletion layer narrows significantly.
- Once the applied voltage exceeds the knee voltage ($V_k \approx 0.7$ V for Si), majority carriers easily cross the junction, resulting in exponential current flow.

3.2 Reverse Bias Mode

When the positive terminal is connected to the n-type region and the negative terminal to the p-type region:

- The external electric field aids the internal built-in field.
- The potential barrier increases to $(V_o + V_r)$, and the depletion layer widens.
- Majority carrier flow is completely blocked. Only a negligible **reverse saturation current (I_o)** flows due to thermally generated minority carriers (microamperes for Ge, nanoamperes for Si).

4. MATHEMATICAL IDEAL DIODE EQUATION

The current-voltage relationship of a PN junction diode under both forward and reverse bias is governed by Shockley's Ideal Diode Equation:

> $$I = I_0 \left[\exp\left(q \cdot V / (\eta \cdot k \cdot T) \right) - 1 \right]$$

Where the parameters are defined as follows:

- **I:** Net diode current
- **I_o:** Reverse saturation current
- **V:** Applied voltage across the diode terminals
- **q:** Absolute charge of an electron (1.602×10^{-19} C)
- **k:** Boltzmann's constant (1.38×10^{-23} J/K)
- **T:** Absolute temperature in Kelvin (K)
- **η:** Ideality factor ($\eta = 1$ for Germanium, $\eta \approx 1$ to 2 for Silicon)
- **$V_T = (k \cdot T / q)$:** Thermal voltage (≈ 26 mV at room temperature, 300 K)

5. V-I CHARACTERISTIC CURVE

The Voltage-Current (V-I) relationship demonstrates clear non-linear conduction characteristics:



F Reverse Voltage V R Forward Current I F (mA) Reverse Current I R (µA) V k (0.7V) V BR (Breakdown)

Figure 2: Representative V-I Characteristic curve for a Silicon PN Junction Diode showing Knee Voltage and Breakdown Region.

6. KEY TECHNICAL SPECIFICATIONS & PARAMETER COMPARISON

Parameter / Property	Silicon (Si) Diode	Germanium (Ge) Diode
Barrier Potential (V_0)	0.7 V	0.3 V
Reverse Saturation Current (I_0)	Very Low (Nanoamperes, nA)	Higher (Microamperes, µA)
Max Operating Temperature	~150 °C – 200 °C	~85 °C
Peak Inverse Voltage (PIV) Rating	High	Moderate to Low
Primary Applications	Power Rectification, Switching	RF Detection, Low Voltage Circuits

7. PRACTICAL APPLICATIONS

1. **Rectification:** Converting alternating current (AC) into direct current (DC) in power supplies using Half-Wave, Full-Wave, and Bridge rectifier circuits.
2. **Clipping & Clamping Circuits:** Modifying wave shapes by removing portions of signal peaks (clippers) or shifting DC voltage reference levels (clampers).
3. **Voltage Regulation:** Special-purpose Zener diodes operate in the reverse breakdown region to maintain a constant output voltage.
4. **Demodulation:** Extracting audio signals from high-frequency radio carrier waves in communication receivers.

8. CONCLUSION

The PN junction diode serves as the fundamental building block of solid-state semiconductor electronics. Its unilateral conductivity property enables crucial functions ranging from power conversion to signal processing. Understanding its barrier dynamics, current transport mechanisms, and temperature dependencies forms the necessary foundation for exploring modern devices such as bipolar junction transistors (BJTs), field-effect transistors (FETs), optoelectronic LEDs, and photovoltaic solar cells.